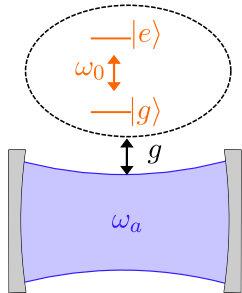


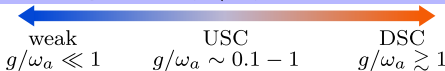
Microscopic picture



Quantum Rabi Model

$$\hat{H}_{RM} = \omega_a \hat{a}^\dagger \hat{a} + \frac{\omega_0}{2} \hat{\sigma}_z + g \hat{\sigma}_x (\hat{a}^\dagger + \hat{a})$$

Coupling intensity (physical orientation)



Effective-theory criterion

$$\frac{\text{relevant coupling matrix element}}{\text{relevant energy separation}} \ll 1$$

typically numerator $\sim g\sqrt{n_* + 1}$

Definitions

$$\Delta = \omega_0 - \omega_a$$

$$\Sigma = \omega_0 + \omega_a$$

n_* = relevant excitation manifold
(e.g., photon number)

Near resonance RWA / Jaynes-Cummings

Criteria:

$$g\sqrt{n_* + 1}/\Sigma \ll 1$$

$$|\Delta|/\Sigma \ll 1$$

Effective Hamiltonian:

$$\hat{H}_{\text{eff}} \propto \hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger$$

RWA: valid

Failure:

$$g\sqrt{n_* + 1}/\Sigma \gtrsim 1$$

or $\delta_{BS} t \gtrsim 1$, $\delta_{BS} \sim g^2/\Sigma$

Key physics:

real 1-excitation exchange
JC doublets /
avoided crossings

Far detuned (Dispersive)

Criteria:

$$g\sqrt{n_* + 1}/|\Delta| \ll 1$$

Effective Hamiltonian:

$$\hat{H}_{\text{eff}} \propto \frac{g^2}{\Delta} (\hat{a}^\dagger \hat{a} \sigma_z + \hat{\sigma}_+ \hat{\sigma}_-)$$

RWA: may apply

Failure:

$$n_* \gtrsim n_{\text{crit}} = \Delta^2 / (4g^2)$$

or $|\delta\phi| \gtrsim 1$

Key physics:

virtual exchange
conditional shifts/phases
approximate QND

Deep strong displaced oscillator (Polaron / GRWA)

Criteria:

$$g/\omega_a \gtrsim 1$$

$$\epsilon_{rs}^{\text{GRWA}} = \frac{|\langle r | \hat{V}_{\text{disc}} | s \rangle|}{|E_r^{\text{ad}} - E_s^{\text{ad}}|} \ll 1$$

Effective Hamiltonian:

$$\hat{H}_{\text{eff}} \propto S_{nn}(g/\omega_a) \hat{\tau}_x^{(n)}$$

RWA: not valid

Failure:

$$\epsilon_{rs}^{\text{GRWA}} \gtrsim 1$$

or remote resonance

Key physics:

displaced-oscillator ladders
paraty doubles

Large-frequency-ratio (LFR)

Criteria:

$$\omega_0/\omega_a \rightarrow \infty$$

$$\lambda = 2g/\sqrt{\omega_0\omega_a} \text{ fixed}$$

Effective Hamiltonian:

$$\lambda < 1: \hat{H}_{\text{eff}} \propto -(\hat{a} + \hat{a}^\dagger)^2$$

$$\lambda > 1: \hat{H}_{\text{eff}} \propto -(\hat{c}_\pm + \hat{c}_\pm^\dagger)^2$$

RWA: not valid

Failure:

$$\omega_0/\omega_a \text{ not large}$$

or λ -scaling not isolated

Key physics:

soft mode / gap closing
normal and superradiant
phases